

Leo C. Stein

CONTACT INFORMATION	205 Lewis Hall University of Mississippi University, MS 38677-1848 USA	lcstein@olemiss.edu duetosymmetry.com 1-662-915-1941
EDUCATION	Ph.D., Physics , Massachusetts Institute of Technology, Cambridge, MA, USA Dissertation Advisor: Prof. Scott Hughes Dissertation Title: <i>Probes of strong-field gravity</i> May 2012 B.S., Physics , California Institute of Technology, Pasadena, CA, USA Degree conferred with honor. Senior Thesis Advisors: Dr. Patrick Sutton and Prof. Alan Weinstein June 2006	
EMPLOYMENT	Associate Professor , University of Mississippi, Oxford, MS USA Assistant Professor , University of Mississippi, Oxford, MS USA Senior Postdoctoral Researcher , Caltech, Pasadena, CA USA NASA Einstein Fellow , Cornell, Ithaca NY, USA Research and Teaching Assistant , MIT, Cambridge MA, USA Teaching Assistant , Caltech, Pasadena, CA, USA Summer Research Fellow , Caltech, Pasadena, CA, USA	July 2024–Present August 2018–June 2024 September 2015–August 2018 September 2012–August 2015 September 2006–May 2012 Fall 2004, Spring 2005 June–September 2003/2005
RESEARCH INTERESTS	General relativity (GR), gravitation, and astrophysical phenomena which can elucidate gravity. One major theme is pushing numerical and analytical gravitational-wave (GW) predictions to the precision frontier in advance of next-generation observatories. A second major theme is using GWs to test GR against beyond-GR models, in both theory-independent and theory-dependent models. This involves numerical relativity and renormalization methods applied to specific effective field models for beyond-GR theories.	
HONORS AND AWARDS	Sloan Research Fellowship , Alfred P. Sloan Foundation, CAREER Award , NSF Einstein Postdoctoral Fellow , NASA Henry Kendall Teaching Award , Massachusetts Institute of Technology Upperclass Merit Scholarship , California Institute of Technology	2023–2025 2021–2026 2012–2015 2011 2005–2006
TEACHING EXPERIENCE	Professor , University of Mississippi Phys. 213, General physics I Phys. 401, Electromagnetism I Phys. 402, Electromagnetism II	Spring 2021 Falls 2019–2022 Springs 2019–2021

Phys. 436, Intro to cosmology	Fall 2023
Phys. 463/4, Senior research project	Fall 2020, Spring 2021, Fall 2023
Phys. 503/630, Graduate reading course	Spring 2019, Falls 2020–2021
Phys. 709, Graduate classical dynamics I	Fall 2018
Phys. 721, Graduate electrodynamics I	Springs 2022–2024
Phys. 722, Graduate electrodynamics II	Falls 2022–2024
Phys. 735, General relativity	Fall 2024
Phys. 750, General relativity II	Spring 2020

Guest Lecturer, California Institute of Technology

Ph236, General relativity	Fall 2017
Ph237, Gravitational Waves	Spring 2016

Guest Lecturer, Massachusetts Institute of Technology

8.901, Graduate Astrophysics I	Spring 2011
--------------------------------	-------------

Teaching Assistant, Massachusetts Institute of Technology

8.942, Cosmology	Fall 2011
8.901, Graduate Astrophysics I	Spring 2011
8.286, The Early Universe	Fall 2009

Teaching Assistant, California Institute of Technology

Ph 7, Nuclear and Quantum Physics Lab	Spring 2005
Ph 5, Analog Electronics for Physicists	Fall 2004

MENTORING/
SUPERVISION**Postdoctoral researchers**

Károly Csukás	Fall 2021–Summer 2024
José Tomás Gálvez Ghersi	Fall 2019–Spring 2021
Now faculty at Universidad de Ingeniería y Tecnología, Peru	

Graduate students

David Bronicki, University of Mississippi	Fall 2019–Summer 2023
Subhayu Bagchi, University of Mississippi	Fall 2019–present
Aniket Khairnar, University of Mississippi	Fall 2019–present
Akshay Khadse, University of Mississippi	Fall 2018–Summer 2024
Lorena Magaña Zertuche, University of Mississippi	Fall 2018–Summer 2024
Now a postdoc at NBI, Copenhagen, Denmark	
Joe Rivest, University of Mississippi	Fall 2018–Summer 2024
Sashwat Tanay, University of Mississippi	Fall 2018–Summer 2022
Now a postdoc at LUTH, Meudon, France	
Maria (Masha) Okounkova, Caltech	Fall 2015–Summer 2019
Now faculty at Pasadena City College	
Baoyi Chen, Caltech	Fall 2016–Summer 2018

PROFESSIONAL
ACTIVITIES,
OUTREACH, AND
SERVICE**Undergraduate students**

Wayne Zhao, Harvard

Summer 2016

Now a graduate student at Princeton

LISA Consortium, Full member

UMiss LISA Group leader

2020–Present**2020–Present****Simulating eXtreme Spacetimes collaboration**

Executive committee member

2015–Present**2018–Present****American Physical Society, member**

Division of Gravitational Physics

2010–Present

Secretary/Treasurer

2023–2026

Executive Committee Member-at-Large

2016–2019

Division of Astrophysics

Conference organizer**Simulating Extreme Spacetimes with SpEC and SpECTRE**, ICERM

Week-long international workshop, ~85 participants

August 2024**New frontiers in strong gravity**, Benasque, Spain

Two week international conference, ~70 participants

July 2024**Nonlinear Aspects of General Relativity**, Princeton PCTS

Four day workshop, ~100 participants

October 2023**Numerical Relativity Community Summer School**, ICERM

Week-long international summer school, 150 participants

August 2022**New frontiers in strong gravity**, Benasque, Spain

Two week international conference, 100 participants

July 2022**Numerical Relativity beyond General Relativity**, Benasque, Spain

Week-long international workshop, 59 participants

June 2018**34th Pacific Coast Gravity Meeting (PCGM)**, Caltech

Two day conference, ~125 participants

March 2018**Unifying Tests of General Relativity**, Caltech

Three day workshop, 52 participants

July 2016**Seminar organizer**

TAPIR seminar, Caltech

Fall 2015–Spring 2018

General Relativity Informal Tea-Time Series (GRITTS), MIT

Fall 2011–Spring 2012

MKI Journal Club, MIT

Fall 2007–Spring 2010**Conference session chair; Judge for best student speaker award**

April APS meeting, NY, NY

April 2022

Midwest relativity meeting, Grand Rapids, MI

October 2019

April APS meeting, Columbus, OH

April 2018**34th Pacific Coast Gravity Meeting (PCGM)**, Caltech**March 2018****33rd Pacific Coast Gravity Meeting (PCGM)**, UCSB**March 2017**

“April” APS meeting, Washington D.C.

January 2017

32nd Pacific Coast Gravity Meeting (PCGM), CSU Fullerton **April 2016**

Theoretical Astrophysics in Southern California (TASC), CSU Fullerton **November 2015**

Journal referee

American Journal of Physics, Classical and Quantum Gravity, Journal of Cosmology and Astroparticle Physics, Journal of Open Source Software, General Relativity and Gravitation, Monthly Notices of the Royal Astronomical Society, Physics Letters B, Physical Review D, Physical Review Letters, Physical Review X, Reviews of Modern Physics, The Astrophysical Journal Letters, The Physics Teacher

Agency work

Reviewer for NSF, NASA

Outreach

Oxford Science Café **April 2019**

Lecture: “The truth about black holes”

Guest on the *Starts With a Bang* podcast **March 25, 2019**

[Episode 42: Black holes and gravitation](#)

Invited speaker for Latin American Webinar on Physics **March 13, 2019**

[Webinar 75: “Testing Einstein with numerical relativity”](#)

Caltech astronomy public lecture series speaker **March 2018**

Lecture: “The truth about black holes”

Astronomy on Tap public lecture series speaker and volunteer **2016–2018**

Close to a monthly basis

Caltech astronomy public lecture series panelist and emcee **2016–2018**

Approximately every three months

Invited guest lecture on black holes and gravitational waves **November 2017**

Science of Space and Time, Hampshire College

Invited video Q&A session, public high school physics class **June 2017**

The Nova Project school, Seattle

Guest on *The Titanium Physicists Podcast*

[Episode 80: Picturing the Bach Hole](#)

August 21, 2019

[Episode 64: The edges of Einstein](#)

April 25, 2016

[Episode 62: Black Bells](#)

February 1, 2016

Quora [Q&A Session](#) on gravitational waves and first detection **February 17, 2016**

83.9k+ views, 20.8k+ followers

Invited guest host, public screening of *COSMOS* with Q&A, **March/June 2014**

Science Cabaret/Cornell

Invited public talk at *Frontiers of Cornell Astronomy*, **November 2013**

Cornell Friends of Astronomy

Invited video chat, *Topics in Physics* course, **July 2013**

Stanford Education Program for Gifted Youth

COMPUTER SKILLS Expert in MATHEMATICA, C/C++, Python, Bash. Proficient in Javascript. Experience in Haskell, Java, Julia. Expert at *nix and HPC. Markup languages: $\text{\LaTeX}$, HTML, CSS, Markdown.

Software—Most contributions can be found at <https://github.com/duetosymmetry>. Member of the *Simulating eXtreme Spacetimes* (SXS) collaboration, contributor to the Spectral Einstein Code (SpEC). Member of the *Black Hole Perturbation Toolkit*. Author of `qnm` python package (<https://github.com/duetosymmetry/qnm>). Core collaborator on xACT (<http://xact.es>) abstract tensor calculus package for MATHEMATICA. Coauthor of xTERIOR package for exterior differential geometry under xACT. Co-maintainer of community contributions at <http://contrib.xact.es>. Developed `arXiv-keys` browser extension/add-on for Chrome/Firefox. Author of `orcidlink` and coauthor of `gridpapers` packages for $\text{\LaTeX}$.

PUBLICATION SUMMARY **h-index** —As of 2024-10-13: 62 (according to Google Scholar), or 55 (according to INSPIRE). Both include collaboration papers.

Top five cited —Excluding LIGO/Virgo collaboration papers.

1. Berti, E., (5 authors), **Stein, L. C.**, (46 more authors) (2015) *Testing General Relativity with Present and Future Astrophysical Observations*, *Class. Quantum Grav.* **32** 243001 [[arXiv:1501.07274](https://arxiv.org/abs/1501.07274)].
2. Barack, L., *et al.* (2019) *Black holes, gravitational waves and fundamental physics: a roadmap*, *Class. Quantum Grav.* **36** 143001 [[arXiv:1806.05195](https://arxiv.org/abs/1806.05195)].
3. Boyle, M., *et al.* (**LCS** is corresponding author) (2019) *The SXS Collaboration catalog of binary black hole simulations*, *Class. Quantum Grav.* **36** 195006 [[arXiv:1904.04831](https://arxiv.org/abs/1904.04831)].
4. Varma, V., *et al.* (2019) *Surrogate models for precessing binary black hole simulations with unequal masses*, *Phys. Rev. Research* **1**, 033015 [[arXiv:1905.09300](https://arxiv.org/abs/1905.09300)].
5. Yunes, N., **Stein, L. C.** (2011), *Nonspinning black holes in alternative theories of gravity*, *Phys. Rev. D* **83** 104002 [[arXiv:1101.2921](https://arxiv.org/abs/1101.2921)].

- SUBMITTED PUBLICATIONS**
64. Witzany, V. Skoupý, V., **Stein, L. C.**, Tanay, S., (2024) *Actions of spinning compact binaries: Spinning particle in Kerr matched to dynamics at 1.5 post-Newtonian order*, [[arXiv:2411.09742](https://arxiv.org/abs/2411.09742)].
 63. Khairnar, A., **Stein, L. C.**, Boyle, M., (2024) *Approximate helical symmetry in compact binaries*, [[arXiv:2410.16373](https://arxiv.org/abs/2410.16373)].
 62. Magaña Zertuche, L., **Stein, L. C.**, *et al.*, (2024) *High-Precision Ringdown Surrogate Model for Non-Precessing Binary Black Holes*, [[arXiv:2408.05300](https://arxiv.org/abs/2408.05300)].
 61. Zhu, H., (9 authors), **Stein, L. C.**, (2024) *Imprints of Changing Mass and Spin on Black Hole Ringdown*, [[arXiv:2404.12424](https://arxiv.org/abs/2404.12424)].

COLLABORATION PUBLICATIONS From 2008–2012, I was coauthor on 34 refereed LIGO and/or LIGO/Virgo collaboration publications. I only list short author-list publications below.

- REFEREED PUBLICATIONS**
60. Sun, D., Boyle, M., Mitman, K., Scheel, M. A., **Stein, L. C.**, Teukolsky, S. A., Varma, V., (2024) *Optimizing post-Newtonian parameters and fixing the BMS frame for numerical-relativity waveform hybridizations*, *Phys. Rev. D* **110** 104076, [[arXiv:2403.10278](https://arxiv.org/abs/2403.10278)].
 59. Mitman, K., Boyle, M., **Stein, L. C.**, *et al.*, (2024) *A Review of Gravitational Memory and BMS Frame Fixing in Numerical Relativity*, *Class. Quantum Grav.* **41** 223001, [[arXiv:2405.08868](https://arxiv.org/abs/2405.08868)].
 58. **Stein, L. C.**, (2024) *Can a radiation gauge be horizon-locking?*, *Class. Quantum Grav.* **41** 157001 [[arXiv:2404.10113](https://arxiv.org/abs/2404.10113)].
 57. Samanta, R., Tanay, S., **Stein, L. C.**, (2023) *Closed-form solutions of spinning, eccentric binary black holes at 1.5 post-Newtonian order*, *Phys. Rev. D* **108**, 124039 [[arXiv:2210.01605](https://arxiv.org/abs/2210.01605)].

56. Bronicki, D., Cárdenas-Avendaño, A., **Stein, L. C.**, (2023) *Tidally-induced nonlinear resonances in EMRIs with an analogue model*, *Class. Quantum Grav.* **40** 215015 [arXiv:2203.08841].
55. Yoo, J., *et al.*, (2023) *Numerical relativity surrogate model with memory effects and post-Newtonian hybridization*, *Phys. Rev. D* **108**, 064027 [arXiv:2306.03148].
54. Ma, S., Varma, V., **Stein, L. C.**, *et al.* (2023) *Numerical simulations of black hole–neutron star mergers in scalar-tensor gravity*, *Phys. Rev. D* **107**, 124051 [arXiv:2304.11836].
53. Tanay, S., **Stein, L. C.**, Cho, G., (2023) *Action-angle variables of a binary black-hole with arbitrary eccentricity, spins, and masses at 1.5 post-Newtonian order*, *Phys. Rev. D* **107**, 103040 [arXiv:2110.15351].
52. Grant, A. M., Saffer, A., **Stein, L. C.**, Tahura, A., (2023) *Gravitational-wave energy and other fluxes in ghost-free bigravity*, *Phys. Rev. D* **107**, 044041 [arXiv:2208.02123].
51. Mitman, K., Lagos, M., **Stein, L. C.**, *et al.* (2023) *Nonlinearities in black hole ringdowns*, *Phys. Rev. Lett.* **130**, 081402 [arXiv:2208.07380]. **§** Editors' Suggestion, **Featured in Physics**.
50. Clark, W. A., Gomes, M. W., Rodriguez-Gonzalez, A., **Stein, L. C.**, Strogatz, S. H., (2023) *Surprises in a classic boundary-layer problem*, *SIAM Review* **2023** 65:1, 291-315 [arXiv:2107.11624].
49. Mitman, K., **Stein, L. C.**, Boyle, M., *et al.* (2022) *Fixing the BMS Frame of Numerical Relativity Waveforms with BMS Charges*, *Phys. Rev. D* **106**, 084029 [arXiv:2208.04356].
48. Okounkova, M., Farr, W. M., Isi, M., **Stein, L. C.**, (2022) *Constraining gravitational wave amplitude birefringence and Chern-Simons gravity with GWTC-2*, *Phys. Rev. D* **106**, 044067 [arXiv:2101.11153].
47. Magaña Zertuche, L., Mitman, K., Khera, N., **Stein, L. C.**, *et al.*, (2022) *High Precision Ringdown Modeling: Multimode Fits and BMS Frames*, *Phys. Rev. D* **105**, 104015 [arXiv:2110.15922].
46. Gálvez Gherzi, J. T., **Stein, L. C.**, (2021) *Numerical renormalization group-based approach to secular perturbation theory*, *Phys. Rev. E* **104**, 034219 [arXiv:2106.08410].
45. Mitman, K., Khera, N., Iozzo, D. A. B., **Stein, L. C.**, *et al.*, (2021) *Fixing the BMS frame of numerical relativity waveforms*, *Phys. Rev. D* **104**, 024051 [arXiv:2105.02300].
44. Iozzo, D. A. B., Khera, N., **Stein, L. C.**, *et al.*, (2021) *Comparing Remnant Properties from Horizon Data and Asymptotic Data in Numerical Relativity*, *Phys. Rev. D* **103**, 124029 [arXiv:2104.07052].
43. Tahura, S., Nichols, D. A., Saffer, A., **Stein, L. C.**, Yagi, K. (2020) *Brans-Dicke theory in Bondi-Sachs form: Asymptotically flat solutions, asymptotic symmetries and gravitational-wave memory effects*, *Phys. Rev. D* **103**, 104026 [arXiv:2007.13799].
42. Tanay, S., **Stein, L. C.**, Gálvez Gherzi, J. T., (2020) *Integrability of eccentric, spinning black hole binaries up to second post-Newtonian order*, *Phys. Rev. D* **103**, 064066 [arXiv:2012.06586].
41. Gálvez Gherzi, J. T., **Stein, L. C.**, (2020) *A fixed point for black hole distributions*, *Class. Quantum Grav.* **38** 045012 [arXiv:2007.11578].
40. Okounkova, M., **Stein, L. C.**, Moxon, J., Scheel, M. A., Teukolsky, S. A., (2020) *Numerical relativity simulation of GW150914 beyond general relativity*, *Phys. Rev. D* **101**, 104016 [arXiv:1911.02588].
39. **Stein, L. C.**, Warburton, N., (2020) *Location of the last stable orbit in Kerr spacetime*, *Phys. Rev. D* **101**, 064007 [arXiv:1912.07609].
38. Okounkova, M., **Stein, L. C.**, Scheel, M. A., Teukolsky, S. A., (2019) *Numerical binary black hole collisions in dynamical Chern-Simons gravity*, *Phys. Rev. D* **100**, 104026 [arXiv:1906.08789].
37. Varma, V., *et al.* (2019) *Surrogate models for precessing binary black hole simulations with unequal masses*, *Phys. Rev. Research* **1**, 033015 [arXiv:1905.09300].

36. **Stein, L. C.**, (2019) `qnm`: A Python package for calculating Kerr quasinormal modes, separation constants, and spherical-spheroidal mixing coefficients, *J. Open Source Softw.*, **4**(42), 1683 [[arXiv:1908.10377](#)].
35. Boyle, M., *et al.* (**LCS** is corresponding author) (2019) *The SXS Collaboration catalog of binary black hole simulations*, *Class. Quantum Grav.* **36** 195006 [[arXiv:1904.04831](#)].
34. Barack, L., *et al.* (2019) *Black holes, gravitational waves and fundamental physics: a roadmap*, *Class. Quantum Grav.* **36** 143001 [[arXiv:1806.05195](#)].
33. Varma, V., **Stein, L. C.**, Gerosa, D., (2019) *The binary black hole explorer: on-the-fly visualizations of precessing binary black holes*, *Class. Quantum Grav.* **36** 095007 [[arXiv:1811.06552](#)], [[project website](#)].
32. Varma, V., Gerosa, D., **Stein, L. C.**, Hébert, F., Zhang, H., (2019) *High-accuracy mass, spin, and recoil predictions of generic black-hole merger remnants*, *Phys. Rev. Lett.* **122**, 011101 [[arXiv:1809.09125](#)].
31. Isi, M., **Stein, L. C.** (2018) *Measuring stochastic gravitational-wave energy beyond general relativity*, *Phys. Rev. D* **98**, 104025 [[arXiv:1807.02123](#)].
30. Prabhu, K., **Stein, L. C.** (2018) *Black hole scalar charge from a topological horizon integral in Einstein-dilaton-Gauss-Bonnet gravity*, *Phys. Rev. D* **98**, 021503(R) (Rapid Communication) [[arXiv:1805.02668](#)].
29. Gerosa, D., Hébert, F., **Stein, L. C.** (2018) *Black-hole kicks from numerical-relativity surrogate models*, *Phys. Rev. D* **97**, 104049 [[arXiv:1802.04276](#)].
28. Chen, B., **Stein, L. C.** (2018) *Deformation of extremal black holes from stringy interactions*, *Phys. Rev. D* **97**, 084012 [[arXiv:1802.02159](#)].
27. Chen, B., **Stein, L. C.** (2017) *Separating metric perturbations in near-horizon extremal Kerr*, *Phys. Rev. D* **96**, 064017 [[arXiv:1707.05319](#)].
26. Okounkova, M., **Stein, L. C.**, Scheel, M. A., Hemberger, D. A. (2017) *Numerical binary black hole mergers in dynamical Chern-Simons: I. Scalar field*, *Phys. Rev. D* **96**, 044020 [[arXiv:1705.07924](#)].
25. Tso, R., Isi, M., Chen, Y., **Stein, L. C.** (2017) *Modeling the Dispersion and Polarization Content of Gravitational Waves for Tests of General Relativity*, *CPT and Lorentz Symmetry*: pp. 205–208 [[arXiv:1608.01284](#)].
24. McNees, R., **Stein, L. C.**, Yunes, N. (2016) *Extremal Black Holes in Dynamical Chern-Simons Gravity*, *Class. Quantum Grav.* **33** 235013 [[arXiv:1512.05453](#)].
23. Flanagan, É. É., Nichols, D. A., **Stein, L. C.**, Vines, J. (2016) *Prescriptions for Measuring and Transporting Local Angular Momenta in General Relativity*, *Phys. Rev. D* **93**, 104007 [[arXiv:1602.01847](#)].
22. Yagi, K., **Stein, L. C.** (2016) *Black Hole Based Tests of General Relativity*, *Class. Quantum Grav.* **33** 054001 [[arXiv:1602.02413](#)].
21. Yagi, K., **Stein, L. C.**, Yunes, N. (2016) *Challenging the Presence of Scalar Charge and Dipolar Radiation in Binary Pulsars*, *Phys. Rev. D* **93** 024010 [[arXiv:1510.02152](#)].
20. Berti, E., (5 authors), **Stein, L. C.**, (46 more authors) (2015) *Testing General Relativity with Present and Future Astrophysical Observations*, *Class. Quantum Grav.* **32** 243001 [[arXiv:1501.07274](#)].
19. Tsang, D., Galley, C. R., **Stein, L. C.**, Turner, A. (2015) “Simplectic” Integrators: Variational Integrators for General Nonconservative Systems, *ApJ* **809** L9 [[arXiv:1506.08443](#)].
18. Yagi, K., **Stein, L. C.**, Pappas, G., Yunes, N., Apostolatos, T. (2014) *Why I-Love-Q: Explaining why universality emerges in compact objects*, *Phys. Rev. D* **90** 063010 [[arXiv:1406.7587](#)].

17. **Stein, L. C.** (2014) *Rapidly rotating black holes in dynamical Chern-Simons gravity: Decoupling limit solutions and breakdown*, **Phys. Rev. D** **90** 044061 [[arXiv:1407.2350](#)].
16. **Stein, L. C.**, Yagi, K., Yunes, N. (2014) *Three-Hair Newtonian Relations for Rotating Stars*, **ApJ** **788** 15 [[arXiv:1312.4532](#)].
15. **Stein, L. C.**, Yagi, K. (2014) *Parameterizing and constraining scalar corrections to general relativity*, **Phys. Rev. D** **89** 044026 [[arXiv:1310.6743](#)].
14. Yagi, K., **Stein, L. C.**, Yunes, N., Tanaka, T. (2013) *Isolated and Binary Neutron Stars in Dynamical Chern-Simons Gravity*, **Phys. Rev. D** **87** 084058 [[arXiv:1302.1918](#)].
13. Yagi, K., **Stein, L. C.**, Yunes, N., Tanaka, T. (2012), *Post-Newtonian, Quasi-Circular Binary Inspirals in Quadratic Modified Gravity*, **Phys. Rev. D** **85** 064022 [[arXiv:1110.5950](#)].
12. Vigeland, S., Yunes, N., **Stein, L. C.** (2011), *Bumpy black holes in alternative theories of gravity*, **Phys. Rev. D** **83** 104027 [[arXiv:1102.3706](#)].
11. Yunes, N., **Stein, L. C.** (2011), *Nonspinning black holes in alternative theories of gravity*, **Phys. Rev. D** **83** 104002 [[arXiv:1101.2921](#)].
10. **Stein, L. C.**, Yunes, N. (2011), *Effective gravitational wave stress-energy tensor in alternative theories of gravity*, **Phys. Rev. D** **83** 064038 [[arXiv:1012.3144](#)].
9. Lutomirski, A., Tegmark, M., Sanchez, N. J., **Stein, L. C.**, Urry, W. L., Zaldarriaga, M. (2011), *Solving the corner-turning problem for large interferometers*, **MNRAS** **410** 2075 [[arXiv:0910.1351](#)].
8. Sutton, P., Jones, G., Chatterji, S., Kalmus, P., Leonor, I., Poprocki, S., Rollins, J., Searle, A., **Stein, L.**, Tinto, M., Was, M. (2010), *X-Pipeline: an analysis package for autonomous gravitational-wave burst searches*, **New J. Phys.** **12** 053034 [[arXiv:0908.3665](#)].
7. Chatterji, S., Lazzarini, A., **Stein, L.**, Sutton, P., Searle, A. (2006), *Coherent network analysis technique for discriminating gravitational-wave bursts from instrumental noise*, **Phys. Rev. D** **74** 082005 [[arXiv:gr-qc/0605002](#)].
6. Galley, C. R., Tsang, D., **Stein, L. C.** (2014) *The principle of stationary nonconservative action for classical mechanics and field theories*, [[arXiv:1412.3082](#)].
5. **Stein, L. C.** (2014), *Note on Legendre decomposition of the Pontryagin density in Kerr*, [[arXiv:1407.0744](#)].
4. **Stein, L. C.** (2012), *Probes of Strong-field Gravity*, Ph.D. thesis at Massachusetts Institute of Technology [[hdl:1721.1/77256](#)].
3. Betancourt, M., **Stein, L. C.** (2011) *The Geometry of Hamiltonian Monte Carlo*, [[arXiv:1112.4118](#)].
2. **Stein, L. C.** (2009), *Binary Inspiral Gravitational Waves from a Post-Newtonian Expansion*, Contribution to the Wolfram Demonstrations Project, <http://demonstrations.wolfram.com/BinaryInspiralGravitationalWavesFromAPostNewtonianExpansion/>
1. **Stein, L. C.** (2006), *Gravitational Wave Burst Source Localization in a Coherent Network Analysis*, Senior thesis at California Institute of Technology

UNREFEREED
PUBLICATIONS

INVITED TALKS

- | | |
|--|----------------|
| 52. UNC physics department colloquium | February 2024 |
| 51. UIUC astrophysics seminar | December 2023 |
| 50. Harvard CMSA GR seminar | October 2023 |
| 49. UMass Amherst, Amherst Center for Fundamental Interactions seminar | September 2023 |
| 48. Albert Einstein Institute, “Connecting the Dots” panel discussion | June 2023 |
| 47. Queen Mary Univ. of London, Gravitational memory workshop | June 2023 |

-
- | | |
|---|------------------------|
| 46. Utah State University, Theoretical Physics Talks, | March 2023 |
| 45. Iowa State, Physics and astronomy department colloquium, | October 2022 |
| 44. UT Austin, Weinberg Institute seminar, | October 2022 |
| 43. Vanderbilt, Physics and astronomy department colloquium, | September 2022 |
| 42. ICERM, Advances in CS Classical and Quantum Gravity, | May 2022 |
| 41. Flatiron CCA, Ringdown workshop, invited overview talk, | February 2022 |
| 40. DAMTP (University of Cambridge), HEP/GR colloquium, | January 2022 |
| 39. SISSA, Current challenges in gravitational physics workshop, | April 2021 |
| 38. Flatiron CCA, Gravitational wave astronomy group seminar, | January 2021 |
| 37. University of Birmingham, astrophysics seminar | September 2020 |
| 36. Albert Einstein Institute, ACR division seminar | July 2020 |
| 35. Black Hole Perturbation Toolkit, Spring 2020 workshop | May 2020 |
| 34. American Physical Society Meeting | April 2020 |
| 33. UVA, physics department colloquium | November 2019 |
| 32. UT Dallas, physics department colloquium | October 2019 |
| 31. Northwestern University, CIERA astrophysics seminar | May 2019 |
| 30. ETH-ITS Zurich, “New horizons for gravity” workshop | May 2018 |
| 29. UC San Diego, astrophysics seminar | March 2018 |
| 28. UC Berkeley, 4D particle physics seminar | March 2018 |
| 27. Kyoto University, YKIS2018a Symposium | February 2018 |
| 26. Oakland University physics seminar | February 2018 |
| 25. University of Wisconsin-Milwaukee gravity seminar | January 2018 |
| 24. Caltech/JPL Gravitational-Wave (CaJAGWR) seminar | January 2018 |
| 23. ICN UNAM, Relativity seminar | December 2017 |
| 22. University of Mississippi, Astrophysics seminar | November 2017 |
| 21. University of Florida, Astrophysics seminar | November 2017 |
| 20. University of Nottingham, Mathematical Physics seminar | July 2017 |
| 19. Sapienza University of Rome, New Frontiers in Gravitational-Wave Astrophysics | June 2017 |
| 18. Rochester Institute of Technology, CCRG seminar | March 2017 |
| 17. Penn State, IGC seminar | March 2017 |
| 16. University of Mississippi, Strong Gravity/Binary Dynamics workshop | February/March 2017 |
| 15. SUNY Stony Brook, “The universe through gravitational waves” | December 2016 |
| 14. University of Pennsylvania, New Frontiers in Gravitational Radiation workshop | December 2016 |
| 13. Cambridge MA, Event Horizon Telescope collaboration meeting | November/December 2016 |
| 12. Northwestern University CIERA, “Fellows at the Frontiers” | August/September 2016 |
| 11. Princeton University, GR@100++ panel discussion | April 2016 |
| 10. Cambridge MA, Einstein fellows symposium | October 2014 |
| 9. Perimeter Institute, Strong gravity seminar | October 2014 |
| 8. Cornell University, Friends of astronomy outreach event | November 2013 |
| 7. Cambridge MA, Einstein fellows symposium | October 2013 |

- | | |
|--|----------------|
| 6. SUNY Geneseo, Physics colloquium | October 2013 |
| 5. University of Maryland, UMD gravity seminar | October 2013 |
| 4. Yale University, YCAA seminar | September 2013 |
| 3. Kyoto University, YITP long-term workshop | June 2013 |
| 2. Cambridge MA, Einstein fellows symposium | October 2012 |
| 1. Cornell University, Relativity lunch | November 2011 |

CONTRIBUTED
TALKS (SELECTED)

- | | |
|---|-------------------------------|
| 23. American Physical Society Meeting | April 2024 |
| 22. American Physical Society Meeting | April 2023 |
| 21. LISA Symposium XIV | July 2022 |
| 20. American Physical Society Meeting | April 2021 |
| 19. American Physical Society Meeting | April 2019 |
| 18. American Physical Society Meeting | April 2018 |
| 17. Pacific Coast Gravity Meeting | March 2017 |
| 16. American Physical Society Meeting | April January 2017 |
| 15. Testing Gravity 2017 | January 2017 |
| 14. 21 st International meeting on GR (GR21) | July 2016 |
| 13. American Physical Society Meeting | April 2016 |
| 12. Eastern Gravity Meeting | May 2015 |
| 11. American Physical Society Meeting | April 2015 |
| 10. NEB 16 Recent developments in gravity | September 2014 |
| 9. American Physical Society Meeting | April 2014 |
| 8. XXVII Texas symposium on relativistic astrophysics | December 2013 |
| 7. 20 th International meeting on GR (GR20) | July 2013 |
| 6. Eastern Gravity Meeting | June 2013 |
| 5. American Physical Society Meeting | April 2013 |
| 4. Caltech TAPIR Seminar | December 2011 |
| 3. Eastern Gravity Meeting | June 2011 |
| 2. American Physical Society Meeting | April 2011 |
| 1. American Physical Society Meeting | April 2010 |

REFERENCES

Scott A. Hughes, Professor of Physics, Massachusetts Institute of Technology
77 Massachusetts Avenue, Bldg. 37-602A
Cambridge, MA 02139
email: sahughes@mit.edu
office phone: 1-617-258-8523

Nico Yunes, Professor of Physics, University of Illinois
249 Loomis Laboratory
1110 West Green Street
Urbana, IL 61801-3003
email: nyunes@illinois.edu
office phone: 1-814-883-2069

Éanna É. Flanagan, Professor of Physics and Astronomy, Cornell University
463 Physical Sciences Building
Ithaca, NY 14853
email: eef3@cornell.edu
office phone: 1-607-255-6534

Yanbei Chen, Professor of Physics, California Institute of Technology
TAPIR 350-17, Caltech
1200 E. California Boulevard
Pasadena, CA 91125
email: yanbei@caltech.edu (please send correspondence to joann@caltech.edu)
office phone: 1-626-395-4258